

Univariate Calculus, Integration, and Series: Introduction to Thermodynamics

1 Introduction to Differentiation

Why do we care about derivatives?

Derivatives are widely used in the natural sciences because they help us understand many phenomena. Some of the most common uses of derivatives are:

- Rate of change: Chemistry deals with many dynamic processes that evolve over time.
- Optimization: Derivatives are widely used to find the minima or maxima of functions.

The derivative of a function $f(x)$ at a point x is the slope of the straight line tangent to $f(x)$ at x . Using a limiting process, the derivative of $f(x)$ at a point x is defined as

$$f'(x) = \frac{df(x)}{dx} = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (1)$$

Fig. 1 illustrates the limiting process as $\Delta x \rightarrow 0$. Eq. 1 is also known as the definition of the **ordinary derivative**.

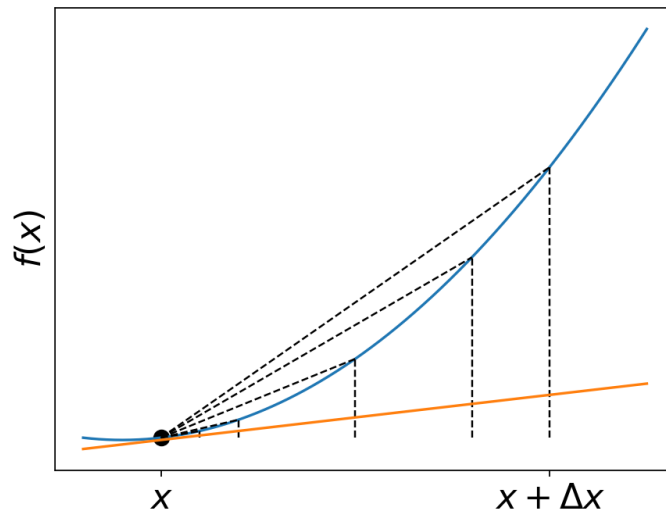


Figure 1: An illustration of the limiting process in the definition of the derivative for $f(x)$.

Link to the animation.

Exercise: Let's compute the derivative of $f(x) = x^2$ using Eq. 1.

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\cancel{x^2} + 2x\Delta x + \Delta x^2 - \cancel{x^2}}{\Delta x} \quad (2)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x(2x + \Delta x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} (2x + \Delta x) = 2x + \lim_{\Delta x \rightarrow 0} \Delta x \quad (3)$$

$$= 2x \quad (4)$$

Exercise: Let's compute the derivative of the so-called **ReLU** (rectified linear unit) function, $f(x) = \max(0, x)$, using Eq. 1. For this exercise, we will consider three scenarios: 1) x is positive, $x \in \mathbb{R}^+$; 2) x is negative, $x \in \mathbb{R}^-$; and 3) $x = 0$.

1. x is positive:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\max(0, x + \Delta x) - \max(0, x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x + \Delta x - x}{\Delta x} = 1. \quad (5)$$

2. x is negative:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\max(0, x + \Delta x) - \max(0, x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0. \quad (6)$$

3.1 x is equal to zero and $\Delta x \rightarrow 0^+$:

$$f'_+(0) = \lim_{\Delta x \rightarrow 0^+} \frac{\max(0, \Delta x) - \max(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0^+} \frac{\Delta x}{\Delta x} = 1. \quad (7)$$

3.2 x is equal to zero and $\Delta x \rightarrow 0^-$:

$$f'_-(0) = \lim_{\Delta x \rightarrow 0^-} \frac{\max(0, \Delta x) - \max(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0^-} \frac{0}{\Delta x} = 0. \quad (8)$$

As we can observe, the right- and left-hand derivatives at $x = 0$ have different values:

$$\begin{cases} f'_+(0) = 1, \\ f'_-(0) = 0. \end{cases} \quad (9)$$

Therefore, the derivative $f'(0)$ does not exist. The ReLU function is **continuous** at $x = 0$, but it is not differentiable there.

For an infinitesimal change dx , the corresponding differential dy is defined as

$$dy = f'(x)dx, \quad (10)$$

where dy and dx are known as the **differentials**.

2 Common derivatives

In practice, one usually consults a table or memorizes the most common derivatives. Later in the course, we will see that derivatives of more complicated functions can be simplified using the **chain rule**. For the following derivatives, we assume that a is a constant and that $f(x)$ and $g(x)$ are both functions of x .

1. Derivative of a constant

$$\frac{da}{dx} = 0 \quad (11)$$

2. Derivative of a sum

$$\frac{d[f(x) + g(x)]}{dx} = \frac{d f(x)}{dx} + \frac{d g(x)}{dx} \quad (12)$$

3. Derivative of a product

$$\frac{d[f(x)g(x)]}{dx} = \frac{d f(x)}{dx}g(x) + f(x)\frac{d g(x)}{dx} \quad (13)$$

4. Derivative of a polynomial

$$\frac{d x^n}{dx} = nx^{n-1} \quad (14)$$

5. Derivative of trigonometric functions

$$\frac{d \cos(x)}{dx} = -\sin(x) \quad (15)$$

$$\frac{d \sin(x)}{dx} = \cos(x) \quad (16)$$

6. Derivative of logarithm

$$\frac{d \ln(x)}{dx} = \frac{1}{x} \quad (17)$$

7. Derivative of exponential

$$\frac{d e^x}{dx} = e^x \quad (18)$$

Using the preceding list of derivatives, one can derive the following:

1. Derivative of $af(x)$

$$\frac{d af(x)}{dx} = a \frac{d f(x)}{dx} + f(x) \cancel{\frac{d a}{dx}} = a \frac{d f(x)}{dx} \quad (19)$$

2. Derivative of the quotient of two functions

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{d}{dx} [f(x)g(x)^{-1}] = g(x)^{-1} \frac{d f(x)}{dx} + f(x) \underbrace{\frac{d g(x)^{-1}}{dx}}_{\text{chain rule}} \quad (20)$$

$$= g(x)^{-1} \frac{d f(x)}{dx} + f(x)(-1)g(x)^{-2} \frac{d g(x)}{dx} \quad (21)$$

$$= \left(\frac{g(x)}{g(x)} \right) \frac{\frac{d f(x)}{dx}}{g(x)} - \frac{f(x) \frac{d g(x)}{dx}}{g(x)^2} = \frac{g(x) \frac{d f(x)}{dx} - f(x) \frac{d g(x)}{dx}}{g(x)^2} \quad (22)$$

$$= \frac{1}{g(x)} \frac{d f(x)}{dx} - \frac{f(x)}{g(x)^2} \frac{d g(x)}{dx} \quad (23)$$

2.1 Chain rule: derivative of a composite function

As you will see throughout this course, it is sometimes convenient to view more complicated functions as compositions of simple functions, $(f \circ g)(x) = f(g(x))$. For example,

$$f(x) = (x+1)^2 \quad (24)$$

$$f(u) = u^2, \text{ where we defined } u(x) = x+1. \quad (25)$$

The derivatives of $f(u)$ with respect to u and of $u(x)$ with respect to x are simpler than the derivative of $f(x)$ with respect to x . We can compute $\frac{df(x)}{dx}$ using the chain rule¹,

$$\frac{df(x)}{dx} = \frac{df(g(x))}{dx} = \frac{df(u)}{du} \times \frac{du}{dx}, \quad (26)$$

where, for convenience, we define the variable $u = g(x)$.

Exercise: Compute the derivative of $f(x) = (x+1)^2$ using the chain rule.

$$\frac{df(x)}{dx} = \frac{du}{dx} \times \frac{df(u)}{du} = (1) \times (2u) = 2(x+1) = 2x+2. \quad (27)$$

Exercise: Let's compute the derivative of $f(x) = e^{-(x^2+a^2)^{\frac{1}{2}}}$ using the chain rule multiple times.

First, we define $u_1 = -(x^2 + a^2)^{\frac{1}{2}}$:

$$f'(x) = \frac{df(u_1)}{du_1} \frac{du_1}{dx} = \frac{de^{u_1}}{du_1} \underbrace{\frac{du_1}{dx}}_{\text{chain rule}} = e^{u_1} \frac{du_1}{dx}. \quad (28)$$

Similarly, we define a second auxiliary function, $u_2 = x^2 + a^2$, so that $u_1 = -u_2^{\frac{1}{2}}$. Because of this relation, $f(x)$ can be decomposed as

$$f(x) = f(u_1) = f(u_1(u_2(x))) = (f \circ u_1 \circ u_2)(x) \quad (29)$$

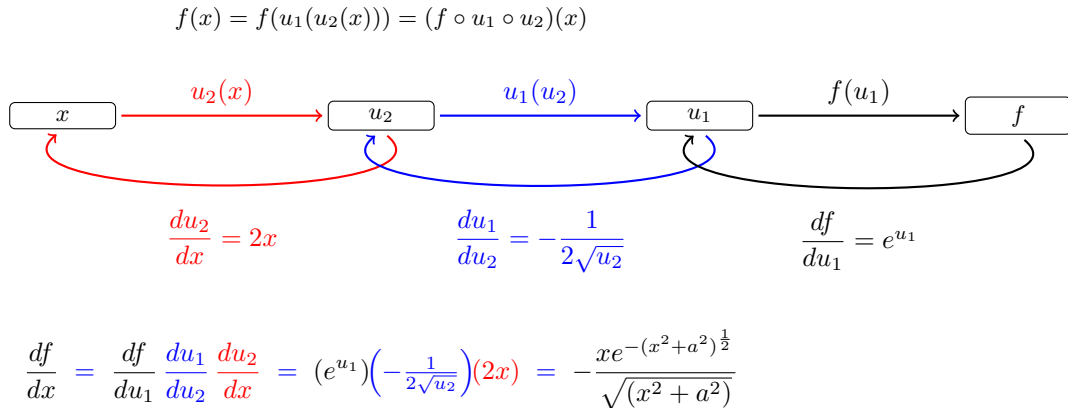
and its derivative is

$$f'(x) = \left(\frac{df(u_1)}{du_1} \right) \left(\frac{du_1}{du_2} \right) \left(\frac{du_2}{dx} \right) = (e^{u_1}) \left(-\frac{1}{2\sqrt{u_2}} \right) (2x), \quad (30)$$

Simplifying this expression and substituting the definitions of u_1 and u_2 , we obtain

$$f'(x) = -\frac{xe^{-(x^2+a^2)^{\frac{1}{2}}}}{\sqrt{x^2+a^2}}. \quad (31)$$

Here is the chain-rule diagram for $f'(x)$:



¹The chain rule is one of the main components of modern machine-learning libraries; we will discuss it further throughout the course.

3 Differentiation for multi-dimensional functions

3.1 Partial Derivatives

In natural sciences, we commonly need functions that depend on multiple variables to describe a phenomenon. For example, in thermodynamics, an equation of state relates temperature T , pressure P , and volume V through a function such as $f(P, V, T)$. Similarly, we can define the derivative of a multivariable function with respect to (w.r.t.) one of its variables. This is known as a **partial derivative**, and it is defined as

$$\left(\frac{\partial f(x, y)}{\partial x} \right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}, \quad (32)$$

where $()_y$ means that the variable y is held constant. We commonly omit the subscript notation $()_y$ because it becomes cumbersome for functions with many variables. Thus, we write $\frac{\partial f(x, y)}{\partial x} = \left(\frac{\partial f(x, y)}{\partial x} \right)_y$.

Eq. 32 is analogous to the definition of the ordinary derivative (Eq. 1). The only difference is the notation used: $\frac{d}{dx} \rightarrow \frac{\partial}{\partial x}$. We can also consider the partial derivative of $f(x, y)$ with respect to y :

$$\left(\frac{\partial f(x, y)}{\partial y} \right)_x = \frac{\partial f(x, y)}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}. \quad (33)$$

The rules for ordinary derivatives also apply to partial derivatives.

Exercise: Let's compute the partial derivatives of $f(x, y) = ax^2y^3 + be^{xy}$ with respect to x and y .

1. w.r.t. x

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial}{\partial x} ax^2y^3 + \frac{\partial}{\partial x} be^{xy} = ay^3 \frac{\partial}{\partial x} x^2 + b \frac{\partial}{\partial x} e^{xy} \quad (34)$$

$$= ay^3(2x) + b ye^{xy} = 2axy^3 + b ye^{xy} \quad (35)$$

$$(36)$$

2. w.r.t. y

$$\frac{\partial f(x, y)}{\partial y} = \frac{\partial}{\partial y} ax^2y^3 + \frac{\partial}{\partial y} be^{xy} = ax^2 \frac{\partial}{\partial y} y^3 + b \frac{\partial}{\partial y} e^{xy} \quad (37)$$

$$= ax^2(3y^2) + b xe^{xy} = 3ax^2y^2 + b xe^{xy} \quad (38)$$

$$(39)$$

• Observe that $\frac{\partial f(x, y)}{\partial x} \neq \frac{\partial f(x, y)}{\partial y}$.

If the variables x and y undergo infinitesimal changes, we can define the **total differential** of $f(x, y)$ as

$$df(x, y) = \left(\frac{\partial f(x, y)}{\partial x} \right)_y dx + \left(\frac{\partial f(x, y)}{\partial y} \right)_x dy. \quad (40)$$

Eq. 40 often appears in thermodynamics and metrology.

3.2 Partial derivative identities

Here we will derive two identities for partial derivatives: i) the **reciprocity relation** and ii) the **cyclic relation**. These two relations are widely used in thermodynamics; see Bridgman's thermodynamic equations. The following derivations and examples are based on the triple product rule and the exact differential.

Suppose that the variables x , y , and z are related by a differentiable equation. Then the following total differentials (Eq. 40) exist:

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz; \quad \text{meaning} \quad x = f(y, z) \quad (41)$$

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy; \quad \text{meaning} \quad z = f(x, y) \quad (42)$$

Substituting the expression for dx into the expression for dz and rearranging, we obtain

$$dz = \left(\frac{\partial z}{\partial x}\right)_y \underbrace{\left[\left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz\right]}_{dx} + \left(\frac{\partial z}{\partial y}\right)_x dy \quad (43)$$

$$dz - \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial z}\right)_y dz = \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial z}{\partial y}\right)_x dy \quad (44)$$

$$\left[1 - \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial z}\right)_y\right] dz = \left[\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z + \left(\frac{\partial z}{\partial y}\right)_x\right] dy \quad (45)$$

The left- and right-hand sides depend on dz and dy , respectively. Since y and z are independent variables, dy and dz may be chosen without restriction. For this last equation to be valid in general, the bracketed terms must be equal to zero.

Reciprocity relation: Let's start with the bracket on the left-hand side of Eq. 45.

$$\left[1 - \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial z}\right)_y\right] = 0 \quad (46)$$

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial z}\right)_y = 1 \quad (47)$$

Rearranging this equation gives the reciprocity relation:

$$\left(\frac{\partial z}{\partial x}\right)_y = \frac{1}{\left(\frac{\partial x}{\partial z}\right)_y} \quad (48)$$

• The reciprocity relation shows that partial derivatives obtained by interchanging the dependent and independent variables are reciprocals of one another.

Example: Later in the course, we will review the ideal gas law, which relates pressure (P), volume (V), temperature (T), and number of moles (n): $PV = nRT$. We can write

$$T = f(P, V) = \frac{PV}{nR}, \quad \left(\frac{\partial T}{\partial P}\right)_V = \frac{V}{nR} \quad (49)$$

$$P = f(V, T) = \frac{nRT}{V}, \quad \left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V} \quad (50)$$

$$(51)$$

From these two equations, we can see that

$$\left(\frac{\partial T}{\partial P}\right)_V \left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{V}{nR}\right) \left(\frac{nR}{V}\right) = 1, \quad (52)$$

so the reciprocity relation holds for $PV = nRT$.

Cyclic relation: Let's start with the bracket on the right-hand side of Eq. 45.

$$\left[\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z + \left(\frac{\partial z}{\partial y}\right)_x \right] = 0 \quad (53)$$

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = - \left(\frac{\partial z}{\partial y}\right)_x. \quad (54)$$

Using the reciprocity relation, Eq. 48, we obtain

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = - \frac{1}{\left(\frac{\partial y}{\partial z}\right)_x} \quad (55)$$

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x = -1. \quad (56)$$

Eq. 56 is known as the cyclic relation.

• The cyclic relation, also known as the **triple product rule** or **cyclic chain rule**, relates partial derivatives of three interdependent variables.

Example: Later in the course, we will review the ideal gas law, which relates pressure (P), volume (V), temperature (T), and number of moles (n): $PV = nRT$. We can write

$$T = f(P, V) = \frac{PV}{nR}, \quad \left(\frac{\partial T}{\partial P}\right)_V = \frac{V}{nR} \quad (57)$$

$$P = f(V, T) = \frac{nRT}{V}, \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT}{V^2} \quad (58)$$

$$V = f(P, T) = \frac{nRT}{P}, \quad \left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P} \quad (59)$$

$$(60)$$

Substituting these three expressions into Eq. 56 gives

$$\left(\frac{\partial T}{\partial P}\right)_V \left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P = \left(\frac{V}{nR}\right) \left(-\frac{nRT}{V^2}\right) \left(\frac{nR}{P}\right) \quad (61)$$

$$\left(\frac{\partial T}{\partial P}\right)_V \left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P = - \left(\frac{T}{V}\right) \left(\frac{nR}{P}\right) = -\frac{nRT}{VP} = -1. \quad (62)$$

We used the ideal gas law, $PV = nRT$, in the last step.

4 Higher-order Derivatives

Functions can also have higher-order derivatives. For example, acceleration is the second derivative of position with respect to time. $f(x, y)$ has four different second partial derivatives. Let's first consider the partial derivatives w.r.t. x and y of $\frac{\partial f(x, y)}{\partial x}$; this partial derivative is also a function of x and y .

$$\frac{\partial}{\partial x} \frac{\partial f(x, y)}{\partial x} = \frac{\partial^2 f(x, y)}{\partial x^2} \quad \text{and} \quad \frac{\partial}{\partial y} \frac{\partial f(x, y)}{\partial x} = \frac{\partial^2 f(x, y)}{\partial y \partial x} \quad (63)$$

The partial derivatives of $\frac{\partial f(x,y)}{\partial y}$ w.r.t. x and y are

$$\frac{\partial}{\partial x} \frac{\partial f(x,y)}{\partial y} = \frac{\partial^2 f(x,y)}{\partial x \partial y} \quad \text{and} \quad \frac{\partial}{\partial y} \frac{\partial f(x,y)}{\partial y} = \frac{\partial^2 f(x,y)}{\partial y^2} \quad (64)$$

Later in the course, we will see that the second-order derivatives of a multivariate function can be arranged in a matrix known as the **Hessian**:

$$\mathbf{H}_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial y \partial x} \\ \frac{\partial^2 f(x,y)}{\partial x \partial y} & \frac{\partial^2 f(x,y)}{\partial y^2} \end{pmatrix} \quad (65)$$

Exercise: Let's compute the Hessian of $f(x,y) = ax^2y^3 + be^{xy}$,

$$\frac{\partial^2 f(x,y)}{\partial x^2} = \frac{\partial}{\partial x} (2axy^3 + b ye^{xy}) = 2ay^3 + by^2 e^{xy} \quad (66)$$

$$\frac{\partial^2 f(x,y)}{\partial y \partial x} = \frac{\partial}{\partial y} (2axy^3 + b ye^{xy}) = 6axy^2 + be^{xy}(1 + xy) \quad (67)$$

$$\frac{\partial^2 f(x,y)}{\partial x \partial y} = 6axy^2 + be^{xy}(1 + xy) \quad (68)$$

$$\frac{\partial^2 f(x,y)}{\partial y^2} = 6ax^2y + bx^2 e^{xy} \quad (69)$$

From the preceding equations, we can see that $\frac{\partial^2 f(x,y)}{\partial y \partial x} = \frac{\partial^2 f(x,y)}{\partial x \partial y}$. Under the appropriate continuity conditions, this equality of mixed partial derivatives is known as *Clairaut's theorem*.

Proof sketch: When the second partial derivatives are continuous, $\frac{\partial}{\partial x} \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \frac{\partial f}{\partial x}$, as suggested by the following difference quotients:

$$\frac{\partial}{\partial x} \frac{\partial f}{\partial y} = \frac{\partial}{\partial x} \underbrace{\left(\lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \right)}_{\text{function}} \quad (70)$$

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \left(\lim_{\Delta y \rightarrow 0} \frac{f(x + \Delta x, y + \Delta y) - f(x + \Delta x, y)}{\Delta x \Delta y} - \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta x \Delta y} \right) \\ &= \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{f(x + \Delta x, y + \Delta y) - f(x + \Delta x, y) - f(x, y + \Delta y) + f(x, y)}{\Delta x \Delta y} \end{aligned}$$

$$\frac{\partial}{\partial y} \frac{\partial f}{\partial x} = \frac{\partial}{\partial y} \underbrace{\left(\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \right)}_{\text{function}} \quad (71)$$

$$\begin{aligned} &= \lim_{\Delta y \rightarrow 0} \left(\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}{\Delta y \Delta x} - \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta y \Delta x} \right) \\ &= \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y) - f(x + \Delta x, y) + f(x, y)}{\Delta y \Delta x} \end{aligned}$$

Example: What is the Hessian of $V(x) = \frac{1}{2}k(x - R_0)^2$?

$$\mathbf{H} = \frac{\partial^2 V(x)}{\partial x^2} = k \quad (72)$$

Example: What is the Hessian of $V(x_1, x_2) = \frac{1}{2}k_1(x_1 - R_0)^2 + \frac{1}{2}k_2(x_2 - R_0)^2$?

$$\mathbf{H} = \begin{pmatrix} \frac{\partial^2 V(x_1, x_2)}{\partial x_1^2} & 0 \\ 0 & \frac{\partial^2 V(x_1, x_2)}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} \quad (73)$$

Force Constants: The constants k_1 and k_2 represent the “stiffness” of the bonds, i.e., how strongly the potential energy increases as the bond lengths deviate from R_0 . Different values of k_1 and k_2 indicate that the two bonds have different stiffnesses.

5 Numerical differentiation

There are cases in which derivatives do not have convenient analytical expressions and must be estimated numerically.

1. A simple two-point approximation using the definition in Eq. 1

$$\frac{df(x)}{dx} \approx \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (74)$$

2. Symmetric difference quotient

$$\frac{df(x)}{dx} \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x} \quad (75)$$

6 Extra exercise

Example: What is the derivative of $f(x)$?

$$f(x) = 10 + \left(\frac{x}{10}\right)^3 e^{\left(\frac{x}{10}\right)^2} \quad (76)$$

We will use the derivative rules for a product, a polynomial, and an exponential, as well as the chain rule.

$$\frac{df(x)}{dx} = \frac{d}{dx} \left(10 + \left(\frac{x}{10}\right)^3 e^{\left(\frac{x}{10}\right)^2} \right) = \frac{d}{dx} \left(\underbrace{\left(\frac{x}{10}\right)^3}_{f_1(x)} \underbrace{e^{\left(\frac{x}{10}\right)^2}}_{f_2(x)} \right) \quad (77)$$

$$\frac{df(x)}{dx} = \left(\frac{df_1(x)}{dx} \right) f_2(x) + f_1(x) \left(\frac{df_2(x)}{dx} \right) \quad (78)$$

First, we will compute the $\frac{df_i(x)}{dx}$ terms.

For $\frac{df_1(x)}{dx}$, we define a “dummy” variable, $u_1(x) = \frac{x}{10}$, where $\frac{du_1}{dx} = \frac{1}{10}$.

$$\frac{df_1(x)}{dx} = \frac{df_1(u_1)}{du_1} \frac{du_1}{dx} = \left(\frac{du_1^3}{du_1} \right) \left(\frac{1}{10} \right) = \frac{3}{10} u_1^2 = \frac{3}{10} \left(\frac{x}{10} \right)^2. \quad (79)$$

For $\frac{df_2(x)}{dx}$, we define a “dummy” variable, $u_2(x) = \left(\frac{x}{10}\right)^2$, where $\frac{du_2}{dx} = \frac{2}{10} \left(\frac{x}{10}\right)$.

$$\frac{df_2(x)}{dx} = \frac{df_2(u_2)}{du_2} \frac{du_2}{dx} = \left(\frac{de^{u_2}}{du_2}\right) \frac{2}{10} \left(\frac{x}{10}\right) = \frac{2}{10} \left(\frac{x}{10}\right) e^{\left(\frac{x}{10}\right)^2}. \quad (80)$$

Combining these equations, we obtain

$$\frac{df(x)}{dx} = \frac{3}{10} \left(\frac{x}{10}\right)^2 e^{\left(\frac{x}{10}\right)^2} + \left(\frac{x}{10}\right)^3 \frac{2}{10} \left(\frac{x}{10}\right) e^{\left(\frac{x}{10}\right)^2} \quad (81)$$

$$\frac{df(x)}{dx} = e^{\left(\frac{x}{10}\right)^2} \left(\frac{3x^2}{1000} + \frac{x^4}{50000} \right). \quad (82)$$